

2.1

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, a, b, c \in \mathbb{R}$$

$$B_{ON} = \left( \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right)$$

$$\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\rangle = 1 \quad \leftarrow \begin{array}{l} \text{bázové vektory} \\ \text{mají normu} = 1 \end{array} \quad \left\langle \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right\rangle = 1$$

$$(a+2b \quad b+2c) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1$$

$$a+2b + 2b+4c = 1$$

$$a+4b+4c = 1$$

$$(3a+7b \quad 3b+7c) \begin{pmatrix} 3 \\ 7 \end{pmatrix} = 1$$

$$9a+42b+49c = 1$$

$$\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right\rangle = 0 \quad \leftarrow \begin{array}{l} \text{bázové vektory} \\ \text{jsou na sebe} \\ \text{v ON kolmé} \end{array}$$

$$(a+2b \quad b+2c) \begin{pmatrix} 3 \\ 7 \end{pmatrix} = 0$$

$$3a+13b+14c = 0$$

$$\det C = \begin{vmatrix} 1 & 4 & 4 \\ 3 & 13 & 14 \\ 9 & 42 & 49 \end{vmatrix} =$$

$$= \begin{vmatrix} 1 & 4 & 0 \\ 3 & 13 & 1 \\ 9 & 42 & 7 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 0 \\ 0 & 1 & 1 \\ 0 & 6 & 7 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

$$\det C_a = \begin{vmatrix} 1 & 4 & 4 \\ 0 & 13 & 14 \\ 1 & 42 & 49 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 6 \\ 0 & 13 & 1 \\ 1 & 42 & 7 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 0 \\ 0 & 13 & 1 \\ 0 & 38 & 7 \end{vmatrix} = 13 \cdot 7 - 38 = 91 - 38 = 53 = a$$

$$\det C_b = \begin{vmatrix} 1 & 1 & 4 \\ 3 & 0 & 14 \\ 9 & 1 & 49 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 4 \\ 0 & -3 & 2 \\ 0 & -8 & 13 \end{vmatrix} = -3 \cdot 13 + 16 = -23 = b$$

$$\det C_c = \begin{vmatrix} 1 & 4 & 1 \\ 3 & 13 & 0 \\ 9 & 42 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 1 \\ 3 & 13 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{vmatrix} = 1 + 9 = 10 = c$$

$$A = \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix} \quad \det A = 1 \quad \wedge \quad \begin{matrix} 53 \\ 10 \end{matrix} > 0 \quad \checkmark$$

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 53x_1y_1 - 23x_1y_2 - 23x_2y_1 + 10y_1y_2$$